

The Late-Time Data's Preference for a Reduced Sound Horizon, and a Locked Phenomenological Modulation that Tracks It

LVC v11 — *Lagrangian Variable Cosmology*, Version 11

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Abstract

In a previous note (v10) we presented a three-parameter modification of the late-time Hubble rate, $H(z) = H_{\Lambda\text{CDM}}(z) \cdot T(z)$, with

$$T(z) = 1 + A \exp \left[- \left(\frac{z - 1/\sqrt{3}}{1/(3\pi)} \right)^2 \right],$$

referred to as *R7*. We argued that *R7* is preferred over ΛCDM on a 36-point combined late-time dataset by $\Delta\text{BIC} = -7.38$. Two reservations were registered there: that *R7*'s preference rests on the data permitting $r_d \approx 137$ Mpc rather than the standard recombination value 147 Mpc, and that the comparison was background-only (no perturbation predictions tested).

In this v11 note we address both points. We do not present a new model. We test whether v10's quantitative claim survives two checks:

- (i) **Fair r_d accounting.** When ΛCDM is allowed the same r_d freedom that *R7* is implicitly using ($k = 3$ for ΛCDM , $k = 4$ for *R7*), *R7* is still preferred by $\Delta\text{BIC} = -6.89$. The two models, fit independently on these data, both prefer $r_d \approx 137$ –139 Mpc. The preference is not an artifact of asymmetric accounting; it is a property of the data.
- (ii) **Joint geometry + $f\sigma_8$.** Adding 19 published $f\sigma_8(z)$ measurements from RSD surveys to the geometric likelihood and computing the linear growth function $D(z)$, $f(z)$ from each model's $H(z)$, *R7* fits $f\sigma_8$ comparably to ΛCDM ($\Delta\chi^2_{f\sigma_8} = +0.21$, marginal). The joint $N = 55$ analysis gives $\Delta\text{BIC}(\text{R7} - \Lambda\text{CDM}) = -6.69$. The growth-of-structure data neither rule out *R7* nor enhance its case.

We report a single negative finding: the data’s preference for $r_d \approx 137$ Mpc, when interpreted via r_d ’s dependence on the relativistic-species number, requires $N_{\text{eff}} \approx 3.8$ – 4.1 for both models, in tension with BBN constraints if N_{eff} is taken literally. We make no attempt to resolve this here. Other mechanisms (early dark energy, varying constants, post-BBN dark radiation) produce the same low- r_d effect without BBN tension and are equally compatible with the data, since the data themselves do not distinguish among them.

The headline numbers $\Delta\text{BIC} = -7.38$ (geometry only, v10), -6.89 (geometry, fair r_d), and -6.69 (geometry + $f\sigma_8$) cluster within 0.7 of one another. This is the strongest claim we can make: the size of R7’s empirical advantage is stable to the two stress tests we have performed. It does not establish R7 as a final theory and it does not address a number of further checks (full Pantheon+ likelihood, full Planck likelihood, DESI Y3) that we have not done.

1. Introduction and scope

The v10 manuscript proposed $T(z)$ above with $z_c = 1/\sqrt{3}$ and $w = 1/(3\pi)$ as locked constants and reported $\Delta\text{BIC} = -7.38$ versus ΛCDM . Two issues, identified there as limitations, are revisited in this note.

First, the v10 fit is consistent with $r_d \approx 137$ Mpc, $\sim 7\%$ below the standard Planck-recombination value of 147.05 Mpc. We did not investigate whether this preference is specific to R7 or whether it is a property of the data common to all reasonable late-time parameterizations. Section 2 of the present note tests this.

Second, R7 is a modification of $H(z)$ and therefore of the linear growth function $D(z)$ for matter perturbations. The v10 likelihood used only background observables (distances). We did not test whether R7 is consistent with $f\sigma_8$ measurements, and a reasonable reviewer would have asked. Section 3 of the present note does this test.

Both checks are constructed to potentially falsify, not to confirm, v10’s claim. We report what we found.

2. Fair sound-horizon comparison

2.1 Setup

In v10 we counted ΛCDM with $k = 2$ (H_0, Ω_m) and R7 with $k = 3$ (H_0, Ω_m, A), with r_d “derived” from $\theta_* \cdot D_M(z_*)$ in both. The θ_* measurement is one likelihood term, so this convention is mathematically equivalent to letting r_d float and being constrained by θ_* . The BIC bookkeeping was symmetric across the two models.

A skeptical reading is that R7 still implicitly absorbs a degree of freedom in r_d that ΛCDM does not exploit, because the R7 fit settles at $r_d \approx 137$ Mpc while ΛCDM with the same convention settles at $r_d \approx 139.5$ Mpc. We test this skepticism by allowing r_d as an explicit free parameter for both models, with θ_* now contributing an explicit χ^2 term.

Under this convention: Λ CDM has $k = 3$ (H_0, Ω_m, r_d); R7 has $k = 4$ (H_0, Ω_m, A, r_d). R7 carries one extra parameter relative to Λ CDM, and BIC penalizes it accordingly.

2.2 Results

Joint best-fit on the 36-point geometric dataset:

Model	k	χ^2	r_d [Mpc]	BIC
Λ CDM, r_d free	3	27.03	139.22	37.78
R7, r_d free	4	16.56	136.87	30.90
Difference (R7 – Λ CDM)	+1	–10.47	–2.35	– 6.89

R7 still wins, by approximately $\Delta\text{BIC} = -6.89$. The reduction from v10's -7.38 to -6.89 reflects the additional $\ln 36 = 3.58$ penalty paid by R7 for explicit r_d freedom, partially offset by the χ^2 improvement that comes from the same. This is the fair comparison.

2.3 The r_d preference is universal

To verify that $r_d \approx 137$ –139 Mpc is preferred by the data themselves, not driven by either model's parameterization, we hold r_d fixed and minimize over the remaining parameters:

r_d [Mpc]	Λ CDM χ^2	R7 χ^2
137	28.06	16.57
140	27.15	18.35
143	29.79	23.20
145	33.34	27.99
147 (Planck)	38.21	33.94

Both models penalize $r_d = 147$. For Λ CDM the penalty is $\Delta\chi^2 = +11.18$ relative to its r_d -free best-fit, equivalent to a 3.3σ tension. For R7 the penalty is $\Delta\chi^2 = +17.38$, or $\sim 4.2\sigma$.

We emphasize what this does *not* show. It does not show that $r_d = 147$ is wrong. The Planck recombination value is determined by CMB observables (full TT, TE, EE spectra) under standard physics, and the data we use here (θ_* , BAO ratios, SH0ES, SN distances) constrain r_d only through their late-time-geometry implications. The data prefer $r_d \approx 137$ on their own terms; whether physical r_d is in fact 137 or 147 cannot be decided here.

2.4 Interpretation via N_{eff}

A common parameterization of how r_d depends on early-universe physics is the relativistic-species count N_{eff} :

$$r_d(N_{\text{eff}}) \approx 147.05 (3.046/N_{\text{eff}})^{0.246} \text{ Mpc},$$

valid in a neighborhood of the Planck cosmology. Inverting this and using the best-fit r_d values from §2.2:

Model	r_d [Mpc]	implied N_{eff}	$N_{\text{eff}} - 3.046$
Λ CDM	139.22	3.80	+0.76
R7	136.87	4.08	+1.03

Both models prefer N_{eff} above the standard 3.046. Both are in tension with BBN, which constrains $N_{\text{eff}} \lesssim 3.4$ at $\sim 1\sigma$. The interpretation as extra relativistic species is therefore not viable for either model on present data.

We list this as a limitation, not as a claim. The r_d - N_{eff} relation is one of several mechanisms that produce a reduced sound horizon. Early dark energy active at $z \sim 3000$, varying fundamental constants, and dark radiation thermalized after BBN all reproduce $r_d \approx 137$ without disturbing BBN. This paper does not have the data to distinguish among them, and we do not advocate for any particular one. We observe only that the empirical preference for low r_d is not unique to R7 and that its mechanism is an open question.

3. Joint analysis with $f\sigma_8$

3.1 Setup

We compute the linear matter growth function $D(z)$ for each model by solving

$$\frac{d^2 D}{d(\ln a)^2} + \left(2 + \frac{d \ln H}{d \ln a}\right) \frac{dD}{d \ln a} - \frac{3}{2} \Omega_m(a) D = 0,$$

with $\Omega_m(a) = \Omega_{m,0}(1+z)^3/E^2(z)$ and $E(z) = H(z)/H_0$. For R7, $E(z)$ includes the locked-Gaussian factor $T(z)$. The growth rate is $f(z) = d \ln D / d \ln a$. We treat the present-day amplitude $\sigma_8(0)$ as a free nuisance parameter; it cannot be predicted by a background-only theory and serves only to set the absolute normalization.

The prediction is

$$f\sigma_8(z) = f(z) \cdot \sigma_8(0) \cdot D(z)/D(0).$$

We compile $N_{f\sigma_8} = 19$ measurements from 6dFGS, SDSS MGS, GAMA, BOSS DR12, WiggleZ, VIPERS PDR-2, eBOSS DR16 (LRG, ELG, QSO, $\text{Ly}\alpha$), and DESI DR1 (LRG1, LRG2, LRG3, ELG2). We treat them as independent Gaussian measurements with the published statistical errors. This is a coarse approximation; the proper RSD analysis uses each survey's full covariance and is beyond the scope of this note.

3.2 $f\sigma_8$ alone

Fit	$\chi^2_{f\sigma_8}$	Ω_m	$\sigma_8(0)$
Λ CDM with Planck values (0.315, 0.811)	11.99	(fixed)	(fixed)
Λ CDM, both free	9.21	0.227	0.845
R7 with v10 best-fit (Ω_m, A), σ_8 free	10.66	0.294	0.825
R7, Ω_m, A, σ_8 free	9.06	0.232	0.848

Two observations.

First, Λ CDM with Planck-CMB-inferred values ($\Omega_m = 0.315$, $\sigma_8 = 0.811$) gives a worse fit ($\chi^2 = 12.0$) than Λ CDM with Ω_m, σ_8 fit to the RSD data ($\chi^2 = 9.2$). The RSD data prefer lower Ω_m and slightly higher $\sigma_8(0)$ than Planck, recovering the well-known S_8 tension. This tension is a feature of the data, not of either model in this paper.

Second, R7 with v10 best-fit values ($\Omega_m = 0.294$, $A = 0.103$) gives $\chi^2_{f\sigma_8} = 10.66$, slightly worse than Λ CDM's RSD-optimal 9.21. The difference is $\Delta\chi^2 = +1.45$. R7 is therefore *not* a better fit than Λ CDM to RSD data alone, but the gap is too small to disfavor either.

3.3 Joint analysis

We add the $f\sigma_8$ likelihood to the v10 geometric likelihood. The joint dataset has $N = 36 + 19 = 55$ points. R7 has $k = 4$ parameters ($H_0, \Omega_m, A, \sigma_8(0)$); Λ CDM has $k = 3$ ($H_0, \Omega_m, \sigma_8(0)$).

Model	k	total χ^2	χ^2_{geom}	$\chi^2_{f\sigma_8}$	BIC
Λ CDM	3	38.30	27.98	10.32	50.32
R7	4	27.60	17.07	10.53	43.63
Difference	+1	−10.70	−10.91	+0.21	− 6.69

R7's joint preference is $\Delta\text{BIC} = -6.69$. Almost all of the χ^2 improvement comes from geometry; in $f\sigma_8$ R7 is essentially neutral relative to Λ CDM. R7's joint best-fit values ($H_0, \Omega_m, A, \sigma_8$) = (73.040, 0.2931, 0.0957, 0.8239) are very close to its v10 geometry-only values, indicating that adding $f\sigma_8$ does not pull the fit substantially.

4. Summary of the three checks

The three reported numbers for R7 versus Λ CDM:

Test	ΔBIC
v10 geometry, r_d derived ($\Lambda\text{CDM } k = 2, \text{R7 } k = 3$)	-7.38
v11 §2 geometry, r_d free ($\Lambda\text{CDM } k = 3, \text{R7 } k = 4$)	-6.89
v11 §3 geometry + $f\sigma_8$ ($\Lambda\text{CDM } k = 3, \text{R7 } k = 4$)	-6.69

The values agree within 0.7 of one another. We read this as: R7’s empirical advantage on these particular datasets is approximately $\Delta\text{BIC} \approx -7$ and is robust against the two specific stress tests of this paper (fair r_d accounting, addition of $f\sigma_8$).

5. Limitations

The list from v10 §8 stands. We restate and update.

1. **Implied early-physics modification.** The data prefer $r_d \approx 137$ Mpc for both ΛCDM and R7 (§2.3). The mechanism is not constrained by these data alone. Interpretation as ΔN_{eff} alone is in tension with BBN (§2.4); other mechanisms (EDE, varying constants, post-BBN thermalized dark sector) are not. We are agnostic about the mechanism and the question is open.
2. **Linear growth, scale-independent only.** Section 3 uses the standard scale-independent linear-growth ODE. It assumes that $T(z)$ multiplies only the background $H(z)$ and does not modify the gravitational constant in the Poisson equation, anisotropic stress, or the matter-dark-energy equivalence principle. A consistent perturbation theory for $T(z)$ has not been written down.
3. **$f\sigma_8$ data treatment is coarse.** We use 19 published mean values with quoted errors as if they were independent Gaussian measurements. The proper analysis convolves each survey’s full likelihood. The $\Delta\chi^2$ between the two models on $f\sigma_8$ is small (+0.21), so the conclusion (R7 $\sim \Lambda\text{CDM}$ on RSD) is unlikely to change qualitatively, but the precise number could shift.
4. **No full Planck likelihood.** We use only θ_* . R7 has not been confronted with the full TT, TE, EE spectra. The $r_d \approx 137$ preference points toward early-universe physics changes that *could* affect those spectra; whether R7 (combined with such a mechanism) is consistent with full Planck has not been tested.
5. **No Pantheon+ unbinned analysis.** Our SN sample is the 13-bin coarse compilation. The fiducial- Ω_m sensitivity test in v10 §5.2 showed the v10 result remains favorable across $\Omega_m^{\text{fid}} \in [0.27, 0.334]$ but at varying strength.
6. **No theoretical derivation of $z_c = 1/\sqrt{3}$, $w = 1/(3\pi)$.** These constants emerge from the data. Their stability under leave-one-out and high-leverage point removal (v10 §5.3) was an empirical observation. We do not have a first-principles derivation. We do not claim one.
7. **Small dataset.** $N = 36$ for geometry, $N = 55$ for joint. DESI Y3, DES Y6, full Pantheon+, full Planck would all change the numbers and possibly the qualitative conclusion.

6. Conclusion

R7’s preference over Λ CDM on this combined dataset is approximately $\Delta\text{BIC} \approx -7$ across three accounting conventions: derived r_d (-7.38), free r_d for both models (-6.89), and joint geometry + $f\sigma_8$ (-6.69). The preference is not eliminated by giving Λ CDM equivalent r_d freedom, and it is not affected materially by the addition of growth-of-structure data. The data prefer $r_d \approx 137$ Mpc independently of which of the two models is used to interpret them.

We do not interpret these results as a confirmation of R7 as a final cosmological model. The preference is a statistical fact about late-time data; the mechanism producing low r_d (relativistic species, early dark energy, modified recombination, post-BBN dark sector, or systematic in the SH0ES / Planck θ_* chain) is not identified by the data. CMB anisotropy spectra, full Pantheon+ likelihood, and DESI Y3 are needed before a stronger claim is appropriate. We defer those to future work and report only what the present checks show: v10’s quantitative claim survives them.

References

(To be completed — DESI DR1 2024, BOSS DR12, eBOSS DR16, Pantheon+, Planck PR4, SH0ES Riess+ 2022, Aylor+ 2019 for the $r_d(N_{\text{eff}})$ relation, plus the $f\sigma_8$ data sources.)

Appendix A: Reproducibility

All numerical results in this paper are reproduced by `lvc_v11_reproduce.py`. The script extends `lvc_v10_reproduce.py` (companion to the v10 manuscript) by adding (a) the r_d -free fits and the r_d profile, and (b) the linear-growth ODE solver, the 19-point $f\sigma_8$ data, and the joint analysis. Tabulated numbers reproduce within ≤ 0.05 in χ^2 and BIC.